

Zhengzhong Ricky You

San Jose, California | +1 (352) 872-8929 | ricky.you.or@gmail.com
[Homepage](#) | [Google Scholar](#) | [LinkedIn](#) | [ORCID](#) | [GitHub](#)

RESEARCH INTERESTS

- Exact algorithms for large-scale integer and combinatorial optimization, with emphasis on branch-price-and-cut, decomposition, and extended formulations.
- Learning-augmented optimization, decision-focused learning, reinforcement learning for combinatorial optimization, and interfaces between operations research and machine learning.
- Large language models for healthcare operations, including routing, scheduling, and resource allocation under uncertainty and fairness constraints.

EDUCATION

- **University of Florida**, M.S. in Operations Research, Industrial & Systems Engineering; GPA: 3.95/4.00 2025
- **Nankai University**, B.S. in Chemistry; GPA: 3.67/4.00 2021

PUBLICATIONS

- [S1] *Learned Pairwise Deep Dual-Optimal Inequalities for Stabilizing Column Generation*.
Zhengzhong Ricky You, Bo Tang, Haoran Liu, Baichuan Mo. Under review at **INFORMS Journal on Computing**, 2026.
- [C1] *Draft-and-Audit Reinforcement Learning for Optimization Modeling*.
Zeping Min, Weihang Xu, Zhengzhong Ricky You, Wotao Yin, Xinshang Wang. **Proceedings of the 43rd International Conference on Machine Learning (ICML 2026)**, PMLR 306, 2026.
- [J1] *RouteOpt: An Open-Source Modular Exact Solver for Vehicle Routing Problems*.
Zhengzhong Ricky You, Yu Yang. **INFORMS Journal on Computing**, 2026.
- [J2] *Two-Stage Learning to Branch in Branch-Price-and-Cut Algorithms for Solving Vehicle Routing Problems Exactly*.
Zhengzhong Ricky You, Yu Yang, Xinshang Wang, Wotao Yin. **Operations Research**, 2026.
- [S2] *Bulk Service Queueing for Transit Resilience under Short Random Service Suspensions*.
Baichuan Mo, Li Jin, Zhengzhong Ricky You, Zuo-Jun Max Shen, Haris N. Koutsopoulos, Jinhua Zhao. Under review at **Transportation Science**, 2016.

RESEARCH EXPERIENCE

- **TikTok Inc.** 2026
Machine Learning Engineer Intern, E-commerce Supply Chain & Logistics San Jose, CA
 - Conducted research on inventory control for e-commerce supply chains under the mentorship of Baichuan Mo, Yikang Hua, and Zhou Ye.
- **MIT-UF-NEU Joint Summer Research Camp** 2026
Selected Participant, MIT JTL-Transit Lab program
 - Studied large-scale inventory placement under demand uncertainty with mentorship from Baichuan Mo.
- **University of Kentucky** 2026
Visiting Scholar, Department of Statistics
 - Conducting research on online learning and branch-and-bound algorithms.
- **DAMO Academy, Alibaba Group** 2025
Research Intern Bellevue, WA
 - Studied large-language-model-assisted solver parameter tuning under the mentorship of Xinshang Wang and Wotao Yin. The tuned configurations improved average benchmark performance over solver defaults; the tested black-box baselines generally did not.

SELECTED RELEVANT COURSES

ISE Ph.D. level (GPA: 4.00): Discrete Optimization; Linear Programming and Network Optimization; Stochastic Modeling & Analysis; Advanced Topics in Continuous Optimization. **Mathematics Ph.D. level (GPA: 4.00):** Analysis I; Numerical Linear Algebra; Numerical Analysis; Numerical Optimization. **CS Ph.D. level (GPA: 4.00):** Machine Learning.

RESEARCH SOFTWARE

- **TorchDCM**  2026
Co-developer, Python Open source
 - Co-developed a unified PyTorch-native package for discrete choice modeling, with vectorized likelihoods, automatic differentiation, multiple logit model families, and standard econometric outputs.
- **RouteOpt**  2021
Lead Developer, 20k+ LOC in C++ Open source
 - Designed and implemented an open-source exact solver for the CVRP and CVRPTW, with modular branching, cutting-plane, and variable-reduction components, learning-augmented branching, and reproducible experiment scripts.
 - On standard benchmarks, RouteOpt nearly doubled the speed of VRPSolver and proved optimality for four previously open instances; one required approximately 1.5 years of single-core time.

CONFERENCE & WORKSHOP PRESENTATIONS

- *L-DDOIs: Learning-Based Deep Dual-Optimal Inequalities for Stabilizing Column Generation*, INFORMS Transportation Science and Logistics Conference, MIT Sloan 2026
- *Two-Stage Learning to Branch in Branch-Price-and-Cut Algorithms for Solving Vehicle Routing Problems Exactly*, invited talk, UK AI and ML Symposium and Nontechnical Workshop 2025
- *L-DDOIs: Learning-Based Deep Dual-Optimal Inequalities for Stabilizing Column Generation*, INFORMS Annual Meeting 2025
- *Accelerating Dynamic Programming via Dual Selection for Column-Generation Frameworks*, INFORMS Optimization Society Conference 2024
- *RouteOpt: A Scalable Advanced Optimization Tool for VRPs*, INFORMS Annual Meeting 2024
- *Two-Stage Learning to Branch in Branch-Price-and-Cut for Exact VRP*, INFORMS Annual Meeting 2023
- *Learning to Branch with Column Generation*, INFORMS Annual Meeting 2022

AWARDS & HONORS

- Harold D. Haldeman, Jr. Graduate Fellowship, University of Florida 2024
- INFORMS Optimization Society Conference Travel Support 2024
- Academic Excellence Scholarship, Nankai University 2020
- Innovation and Entrepreneurship Scholarship, Nankai University 2019
- First Prize, Tianjin Theoretical Chemistry Competition 2019
- First Prize, Tianjin Mathematics Competition 2018

TEACHING & MENTORING

- **Sole Instructor**, ESI 3327C Matrix & Numerical Methods in Systems Engineering, University of Florida; 52 undergraduates; linear algebra, numerical methods, and introductory optimization 2024
- **Outreach Mentor**, UF Student Science Training Program; mentored Ken Cheng and Shrey Gupta

PROFESSIONAL SERVICE

Journal referee for *INFORMS Journal on Computing*, *INFORMS Journal on Optimization*, *Transportation Science*, *Transportation Research Part B*, *IIE Transactions*, *Computers & Operations Research*, and *OR Spectrum*.

SKILLS

Methods: branch-price-and-cut, column generation, dual stabilization, cutting planes, learning-augmented optimization, stochastic and robust optimization. **Programming:** C++, Python (PyTorch and reinforcement-learning libraries), MATLAB, Shell, Git, Linux. **Software:** RouteOpt, TorchDCM, Gurobi, CPLEX, SCIP, reproducible experiment pipelines.